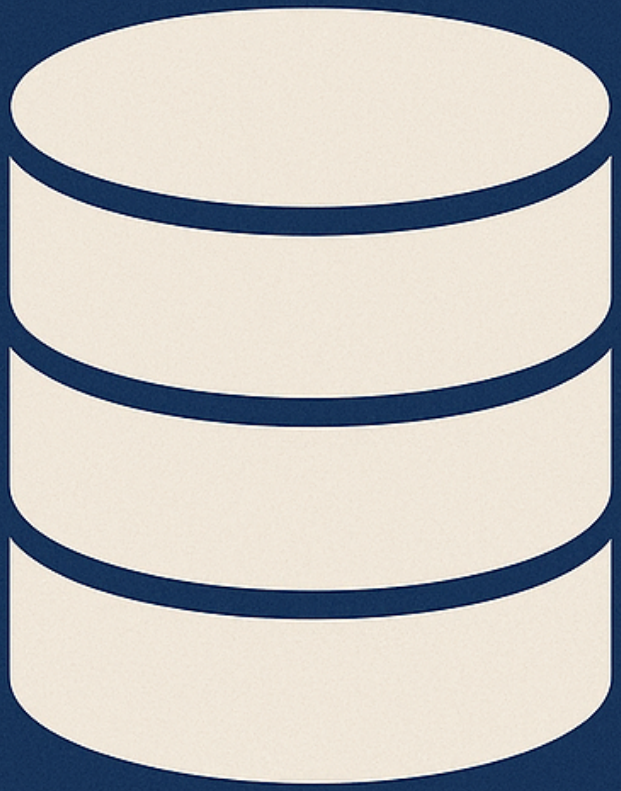
SQL

Class starts
@ 9:10pm





Lecture 8: CTE and UNION

#Agenda

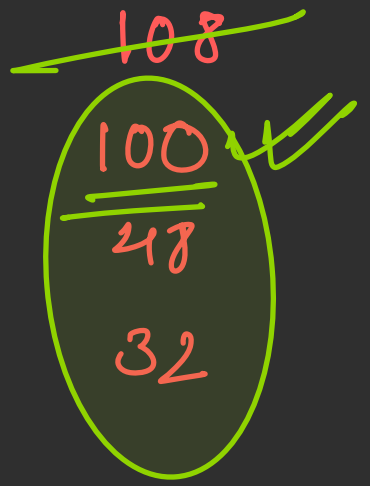
1) Nested Subqueries
2) CTE (Common Table Expression)
3) Union

Nested Subqueries

```
Select col1, col2
From Table
where colX = ( Select Agg_fun (colX) from Table
               where colX < (Select Agg fun (colY)
                             from Table))
```

inner ⟶ outer

user id	team	total mining
1	A	100
2	B	32
3	C	108
4	A	48

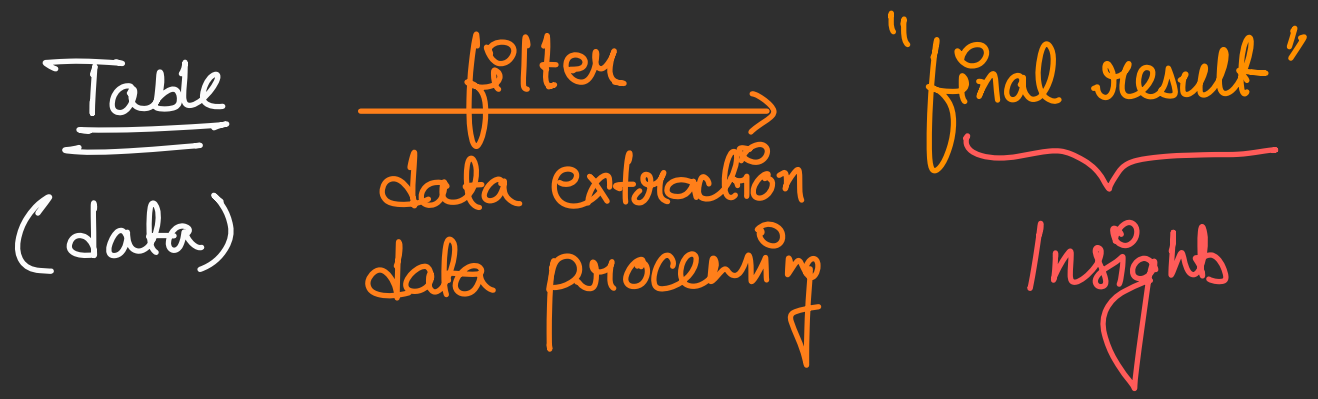


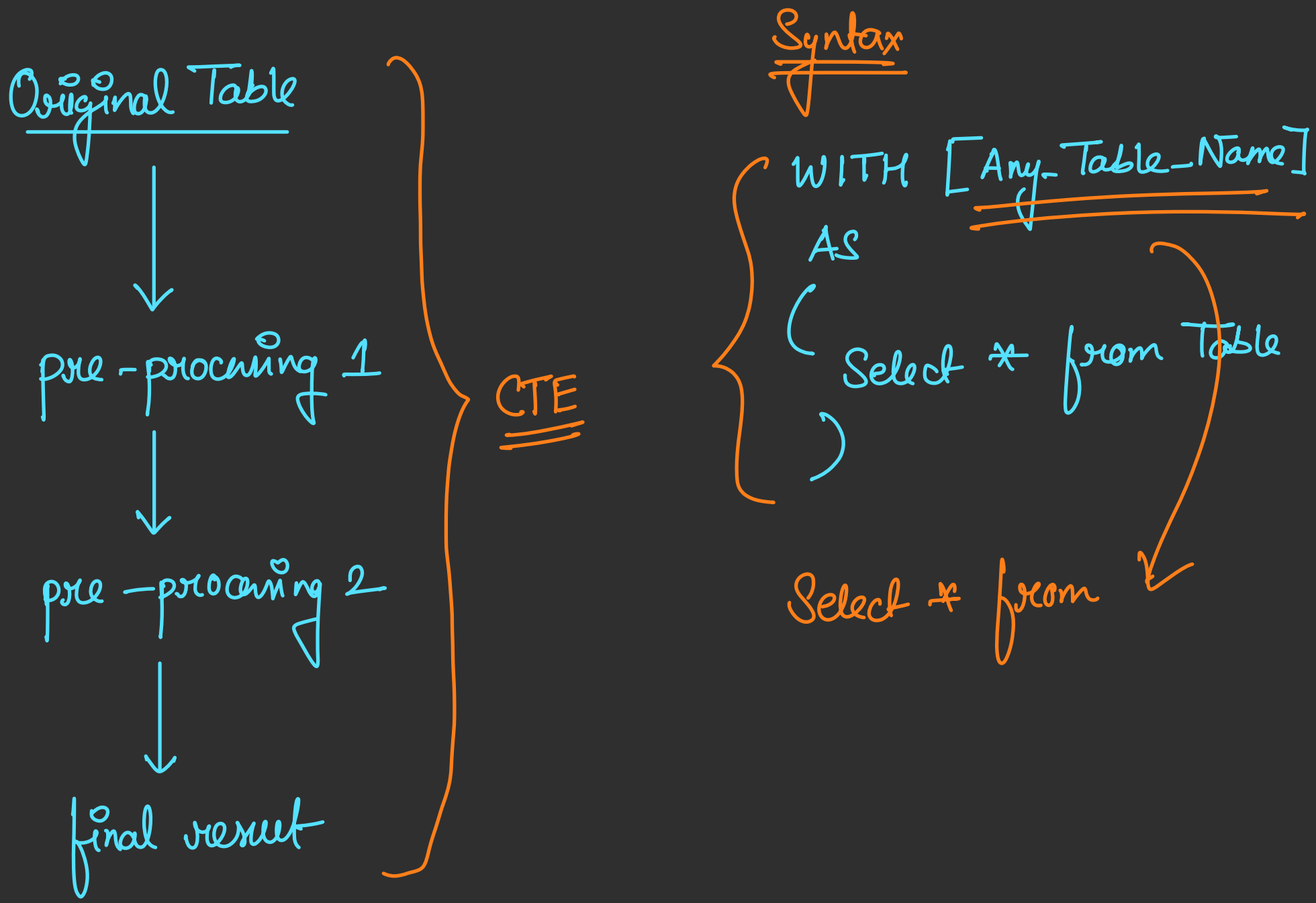
Select Max(total mining) from Table
where total mining <

Select Max(total mining)
from table

→ 108

CTE (Common Table Expression)




Original Table
pre-procsing 1
pre-procsing 2
final result
CTE
Syntax
WITH [Any_Table_Name]
AS
(
Select * from Table
)
Select * from

UNION / UNION ALL

UNION

Table A

col 1	col 2
1	A
2	B
3	C

Table B

col 1	col 2
6	A
1	A
3	X

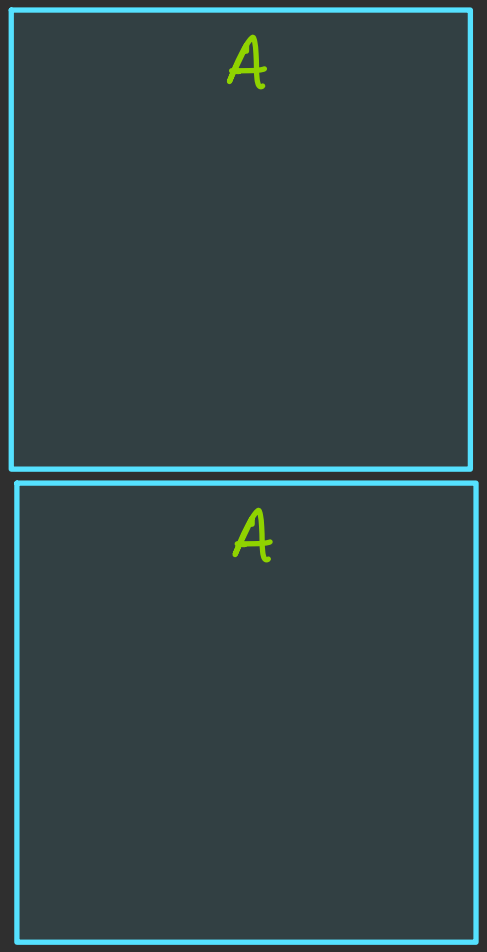
col 1	col 2
1	A
2	B
3	C
6	A
3	X

UNION

col 1	col 2
1	A
2	B
3	C
6	A
3	X

UNION ALL

col 1	col 2
1	A
2	B
3	C
6	A
3	X
1	A



Table

A	1
A	3
A	1

distinct

A	1
A	3

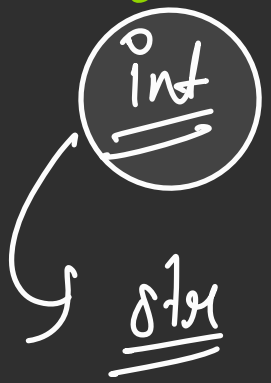
Syntax

str

Select col1, col2 from TableA

UNION / UNION ALL

Select col1, col2 from TableB



MySQL → UNION

Bigquery → UNION DISTINCT